

Non-invasive ventilation (NIV)

A guide to aid appropriate care and disposition of adults who meet any of the criteria listed in box 1

DO NOT apply this pathway if

- NIV is contraindicated due to
 - Severe facial deformity
 - Facial burns
 - Fixed upper airway obstruction
- AHRF is not caused by conditions as per box 1 but entirely due to
 - Overdose
 - Asthma
 - Pneumonia

Disclaimer: This is a clinical template; clinicians should always use judgment when managing individual patients

Re-approved by ED guidelines committee on 25Feb26
Review due: Feb31 - Trust Ref: C185/2016

Patient details

Full name

DoB

S number

(use addressograph if available)

Starting blood gas

Time

pH

pCO₂

Base excess

ABG or ELBG before NIV start

Time

pH

pCO₂

Base excess

ABG or ELBG after 2h on NIV

Time

pH

pCO₂

Base excess

- Record initial blood gas result (VBG acceptable) on the left
- Ensure observations are set to NEWS hypercapnic model
- If SpO₂ <88%, give **CONTROLLED** oxygen (Venturi mask)
- Move to/manage in the resuscitation room (ER)
- Reverse adverse effects of any opiates or benzodiazepines
- Request CXR (but do not delay care while waiting for it)
- Exclude pneumo- or haemothorax, and drain if identified
- If critically unwell (see box 2) **AND** critical care appropriate (see box 3), refer to ICU team **NOW** *

Arterial or **earlobe blood gas** (ELBG) required to confirm (**NOT** VBG)
NB: Electrolyte results on ELBG **NOT** reliable

Use NIV cautiously if

- Copious respiratory secretions (consider respiratory physio; may benefit from cough-assist device)
- Previous ventilator-associated pneumothorax (use lower pressure settings)

* Contact ICU SDM (senior decision maker) on **07971 626 253** or ICU 'registrar' on **Bleep 4458**

- 15min for each meal
- 5-10min up to every 3h for oral hygiene
- **NB:** Restart NIV STAT if respiratory rate >25 or SpO₂ falling

Exclusion include

- AKI 3 or on dialysis
- Rib fractures/trauma
- Pulmonary oedema

ED patients on NIV often face significant delays to ward transfer. Review disposition decisions regularly; may need revising if the patient's condition changes significantly while waiting.

Print name

Signature

Role

Date

① NIV initiation criteria

pH <7.35 and pCO₂ >6.5 in patient with

- AECOPD and respiratory rate >23 ☐
- Acute cardiogenic pulmonary oedema (ACPE) and respiratory rate >23 ☐
- Rib fractures and respiratory rate >23 ☐

Respiratory illness in patient with neuromuscular disease, resp rate > 20 **AND**

- EITHER** pH < 7.35 **AND** pCO₂ >6.5 ☐
- OR** pCO₂ >6.0 if known vital capacity <1L ☐

Obesity hypoventilation syndrome (OHS), respiratory rate >23 **AND**

- EITHER** pH <7.35 **AND** pCO₂ >6.5 ☐
- OR** pCO₂ > 6.0 **AND** daytime drowsiness ☐

② Critical illness criteria

- Imminent or actual respiratory arrest ☐
- SpO₂ <86% even on ≥60% oxygen ☐
- Arterial pH <7.15 ☐
- GCS <9 ☐
- Haemodynamic instability (shock) ☐
- Agitation despite calming medicines ☐
- Intestinal obstruction ☐
- Pneumo- or haemothorax with pH <7.25 ☐
- NIV failing to reduce pCO₂ ☐

③ Critical care appropriate?

☐ **No**, because of the reason(s) below

- Patient with mental capacity declines ☐
- Too frail to benefit from critical care ☐
- Advance Decision to Refuse Treatment (ADRT) or ReSPECT form that sets the ceiling of care below ICU ☐
- NB:** People with a CFS > 5 will usually not benefit from critical care. Those with CFS 5 should be referred for assessment.

☐ **Yes** – as none of the above

④ COPD initial care bundle

To prescribe in NC, go to Emergency Medicine (ED) > Common scenarios (ED) > Asthma & COPD

Tick off each item when prescribed

Nebulizers on AIR (if needed, give nasal spec O₂ to maintain SpO₂ 88-92%)

- ☐ Salbutamol 5mg PRN (back-to-back 1h)
- ☐ Ipratropium 500microgram 2 doses
- Steroids:**
- ☐ Prednisolone 30mg PO **OR** (only if unable to take orals) hydrocortisone 100mg IV
- Antibiotics:**
- ☐ Guided by CXR findings, use the UHL antimicrobial website (Eolas) to treat for pneumonia or LRTI as appropriate

⑤ NIV initiation bundle

Tick off each item when completed

- ☐ Start NIV as per setup guide in box 7
- ☐ Control nausea/agitation/pain if present
- ☐ If new to NIV, offer patient the **NIV PIL**
- ☐ Ensure ReSPECT form including decisions on CPR and escalation to ICU in the event of deterioration or futility is completed
- ☐ Request 'Rapid RSV/influenza virus PCR' swab
- ☐ Start 'NIV acute patient record' and 'NIV audit form' (both provided in NIV tray)
- ☐ Notify DART nurse on **Bleep 5293** or dial **#6826** (links to mobile **07960 779855**)

⑥ Transfer safety criteria met?

☐ **NO**, as one or more of the below

- Deteriorating despite NIV ☐
- Ongoing agitation or vomiting ☐
- Need to control comorbidities, e.g. arrhythmia or abnormal electrolytes ☐
- pH < 7.25 or falling ☐
- SpO₂ despite >60% O₂ < 88% ☐
- GCS < 13 ☐
- Respiratory rate > 29/min ☐
- < 11/min ☐
- Heart rate > 130/min ☐
- < 50/min ☐
- Systolic BP > 90mmHg ☐
- MAP > 130 ☐

☐ **YES**, as none of the above

⑦ NIV setup guide

- Set up NIV only if you are competent in operating the **Philips Trilogy Evo** ventilator
- ED nurses are trained to assemble tubing and fit masks (ER coordinator to help if needed)
- Always start by selecting '**New Patient**'; this will reset the ventilator from previous use
- Escalate IPAP within the suggested time frame to ensure NIV can be maximally effective
- **In pts with previous ventilator-associated pneumothorax, use lower IPAP settings**

Ventilator settings (pressures stated in cmH₂O)

Condition	EPAP *	Initial IPAP	IPAP up-titration over 10-30min	Inspiratory Time (Insp. Time)
	3 (4-8 if OSA)	15 (20 if pH <7.25)	to 20 – 30	1 sec
	5 up to 8	10	may go up to 12 NB: Limit IPAP-EPAP difference to 3-5	1 sec
	3 (4-8 if OSA)	15 (20 if pH <7.25)	to 20 – 30 **	1.2 and also set Insp. T. Min/Max' to 1.2/1.5 sec in 'Advanced'
	3 (4-8 if OSA)	10 (or 5 above their usual)	Not usually required NB: If pCO ₂ not falling, go up to 15	1.2

* Potential need for EPAP >8 (only after RSU/CDU consultant advice)

In severe OHS (BMI >35), to help lung recruitment (e.g. if hypoxic in severe kyphoscoliosis), or to oppose intrinsic PEEP in severe airflow obstruction

NB: To maintain adequate pressure support when high EPAP required, you may also need to increase IPAP

** Potential need for IPAP >30

Might be indicated in OHS

⑧ GH referral process

RSU

ED referrer

- Ensures latest ABG or ELBG is available on NC (either as a Result or as Clinical Photography)
- eRefers to '**ED Referral RSU**' (goes to RSU registrar). In the 'Diagnosis' field, states relevant condition and adds the comment 'on NIV'.
- In the '**ED Bed Requests**' profile, selects
 - In 'ED Specialty Required': 'Respiratory'
 - In 'ED/SDEC Requested Ward': 'GH > Respiratory Support Unit'
 - In 'Bed Request': 'Pending Referral'

Suitability confirmed?

Y

N

RSU registrar

Closes referral with the outcome 'Admit' **AND** changes 'Bed Request' to 'Requested'

RSU NIC

Changes 'Bed Request' to 'Ready' (green) when one of the below:

- Side room free, even if viral PCR result is still pending or positive
- Any bed free if viral PCR negative

Referrer

Ensures an ABG or ELBG has been documented on NC within the last 30min (and obtains one if not)

ED ward clerk

Tasks inter-site transfer ambulance

Transfer on NIV

RSU registrar

- Notifies ED referrer by phone
- Makes appropriate onward referral (this will usually be the CDU team)
- Changes 'ED/SDEC Requested Ward' to 'GH > CDU'
- Closes referral with the outcome 'Referred to other specialty' and adds relevant detail in comments

CDU triage NIC

Checks that RSU registrar has correctly changed the 'ED/SDEC Requested Ward' to 'GH > CDU' **AND** then changes bed request to 'requested'

CDU triage NIC

Changes 'Bed Request' to 'Ready' (green) when bed available

CDU

ED referrer

- Ensures latest ABG or ELBG is available on NC (either as a Result or as Clinical Photography)
- Creates NC '**ED Referral CDU Respiratory**'. In the 'Diagnosis' field, states relevant condition and adds the comment 'on NIV'.
- In the '**ED Bed Requests**' profile, selects
 - In 'ED Specialty Required': 'Respiratory'
 - In 'ED/SDEC Requested Ward': 'GH > CDU'
 - In 'Bed Request': 'Pending Referral'

CDU triage NIC

Closes referral with the outcome 'Admit' **AND** changes 'Bed Request' to 'Requested'

CDU triage NIC

Changes 'Bed Request' to 'Ready' (green) when bed available

Referrer

Ensures an ABG or ELBG has been documented on NC within the last 30min (and obtains one if not)

ED ward clerk

Tasks inter-site transfer ambulance

Transfer on NIV